

Lightweight Fault Detection and Localisation in AWS Serverless Microservices — Accuracy and Monitoring Overhead against Deep-Learning Baselines

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Abstract

Microservice faults propagate across call chains, and top published root-cause detectors train deep models on large labelled corpora—resources a cost-constrained AWS serverless service typically lacks. This study asks what accuracy-overhead position rule-based fault detection and localisation occupy relative to learned deep baselines in AWS serverless microservices. A hybrid lightweight artefact combines CloudWatch 3-sigma detection, optional Isolation Forest localisation, and X-Ray ranking on an order-processing SAM/Terraform stack, evaluated in three legs: citation ceiling versus Xing et al. (2025) $F1 \approx 0.938$; like-for-like RCAEval RE2-OB ($n=90$) against BARO, CIRCA, and TraceRCA; and live AWS lite overhead under full, policy, and off conditions (destroy-after), including Final-3. Leg 2 yields rule-arm detection $F1 = 0.469$ and localisation $AC@3 = 0.611$ versus BARO/CIRCA $AC@3 = 0.878$ —not within 10 pp of the deep ceiling—while hybrid $AC@3 = 0.478$ underperforms rules on Top-3. Final-3 policy-versus-full volume reductions measure $0.383 / 0.422 / 0.472$, so the $\geq 50\%$ limb is not stably met. The joint competitiveness rule fails on accuracy and is not a stable overhead win. CausalRCA full $n=90$ and confirmatory 30-minute cells remain unresolved.

Introduction

Modern cloud-native systems decompose business logic into independently deployable microservices. AWS Lambda and related serverless platforms remove server management and bill per invocation, which makes small, cost-sensitive services attractive—but also makes continuous deep-model training hard to justify. When latency rises or errors spike, operators must localise the respon-

sible service in a distributed call graph. Deep dual-channel and causal RCA methods report high F1 and top-k precision on rich labelled corpora; training-free thresholds and peer baselines offer an alternative when history is absent. For greenfield serverless deployments, choosing observability is a joint accuracy–cost decision: more traces and logs improve localisation odds but raise ingest and tracing spend. Publishing accuracy without overhead, or overhead without common-data accuracy, leaves that decision under-specified. Beneficiaries are SRE and platform teams instrumenting new Lambda services, and researchers comparing untrained versus learned RCA under explicit protocols.

The research question asks what accuracy-overhead position rule-based fault detection and localisation occupy relative to learned deep baselines in AWS serverless microservices. The work designs and implements a lightweight hybrid method—CloudWatch 3-sigma rules, an unsupervised localisation option, and X-Ray ranking—on an AWS serverless order-processing artefact, with success defined by a tested SAM/Terraform stack and detector/ranker code paths. It establishes a three-leg evaluation protocol: citation ceiling against Xing et al., like-for-like RCAEval baselines, and live overhead measurement, requiring Leg 2 tables recomputed from committed raw JSON and Leg 3 overhead.json plus Final-3 packs with destroy verification. Pre-committed expectations are validated as follows: detection F1 within ≤ 10 pp of the strongest comparable baseline; localisation Top-3 may concede more than 10 pp versus deep methods; and policy monitoring reduces telemetry volume by $\geq 50\%$ versus full tracing and logging under the stated sampling policy, with each limb accepted or rejected from evidence files only. The joint decision rule treats the method as competitive only if the accuracy limb and the $\geq 50\%$ volume reduction both hold, reported as a joint boolean with no invented F1-from-Top-3 shortcuts.

Evaluation is controlled. Faults aligned with the RCAEval taxonomy are injected on a four-Lambda topology; rules and hybrid methods are scored offline on RE2-OB (n=90); Xing et al. is cited as ceiling only; live overhead is measured under full, sampled-policy, and off conditions at lite rate and duration; stacks are destroyed after rounds. CausalRCA is attempted but flagged fixed-order and quarantined at n=4. Scope is a single AWS account and region and one microservice topology with synthetic traffic only. Lite Leg 3 windows are not 30-minute confirmatory cells; overhead varies across Final-3 repeats; CausalRCA is not a full peer. External validity does not extend to

multi-cloud estates or production traffic mixes. Working assumptions are that CloudWatch metrics and X-Ray traces are sufficient signals for the committed detectors, that list-price arithmetic on measured volumes estimates dollar cost, that RCAEval case labels are the localisation ground truth for Leg 2, and that destroy-after rounds isolate cost and state.

Chapter 2 reviews learned RCA, training-free detection, benchmarks, and overhead. Chapter 3 details the three-leg methodology and hypotheses. Chapter 4 specifies artefact design (injection, rules, ranking). Chapter 5 covers implementation. Chapter 6 reports Leg 2/3 evidence and joint-rule outcomes. Chapter 7 concludes and lists future work.